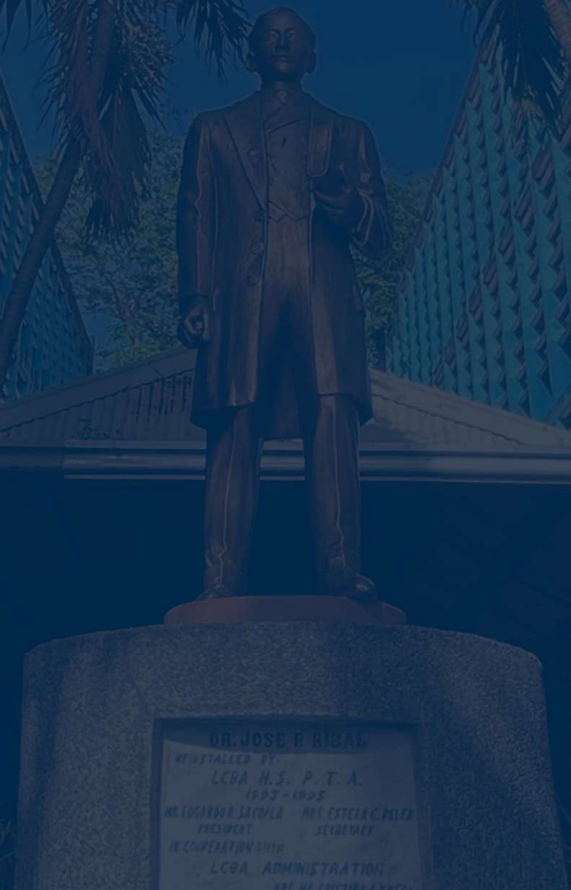




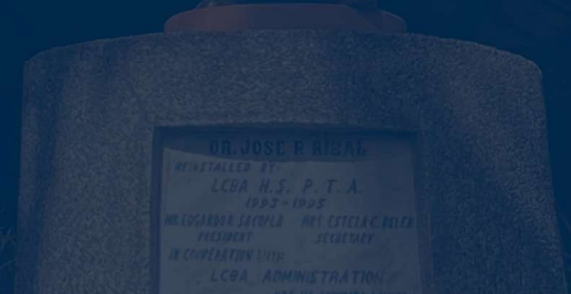
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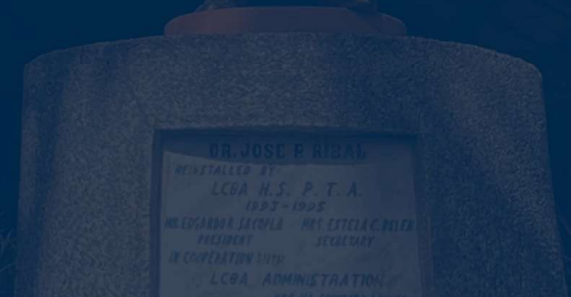
GENERAL MATHEMATICS





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ARITHMETIC AND GEOMETRIC SEQUENCE





Comparing Arithmetic and Geometric Sequences

Feature	Arithmetic Sequence	Geometric Sequence
Growth pattern	Adds same amount each step	Multiplies by same ratio
Common value	Difference <i>d</i>	Ratio <i>r</i>
Example in finance	Fixed savings, salary raises	Compound interest, depreciation
Formula	$a_n = a_1 + (n - 1)d$	$a_n = a_1 \cdot r^{(n-1)}$



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Illustrative Examples 1: Savings and Investments

1. Arithmetic Sequence – Savings:

- Weekly savings: ₱500
- Sequence: 500, 1000, 1500, ...
- After 10 weeks:



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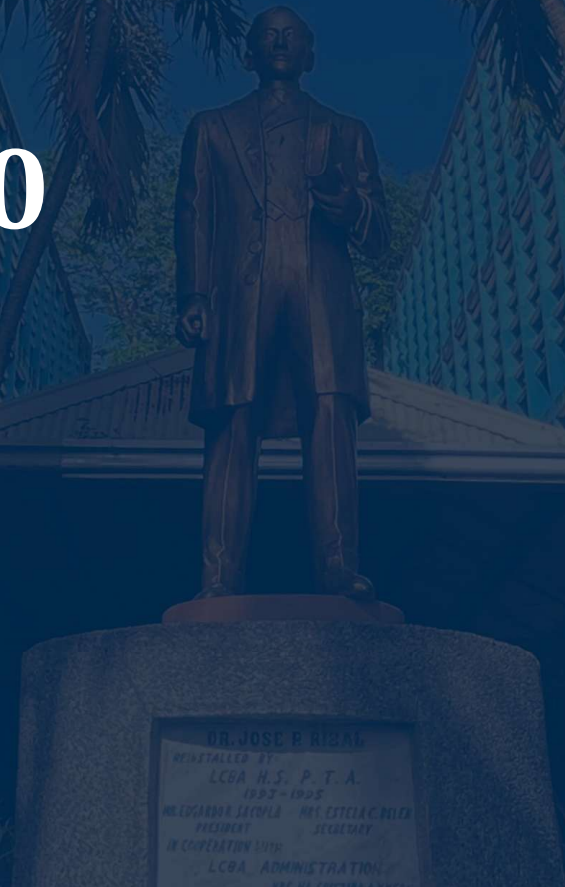
Solution:

$$a_n = a_1 + (n - 1)d$$

$$a_{10} = 500 + (10 - 1)500$$

$$a_{10} = 500 + 4,500$$

$$a_{10} = 5,000$$





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2. Geometric Sequence – Investment:

- Initial investment: ₱20,000
- Monthly growth: 5%
- After 6 months:





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$$a_n = a_1 \cdot r^{(n-1)}$$

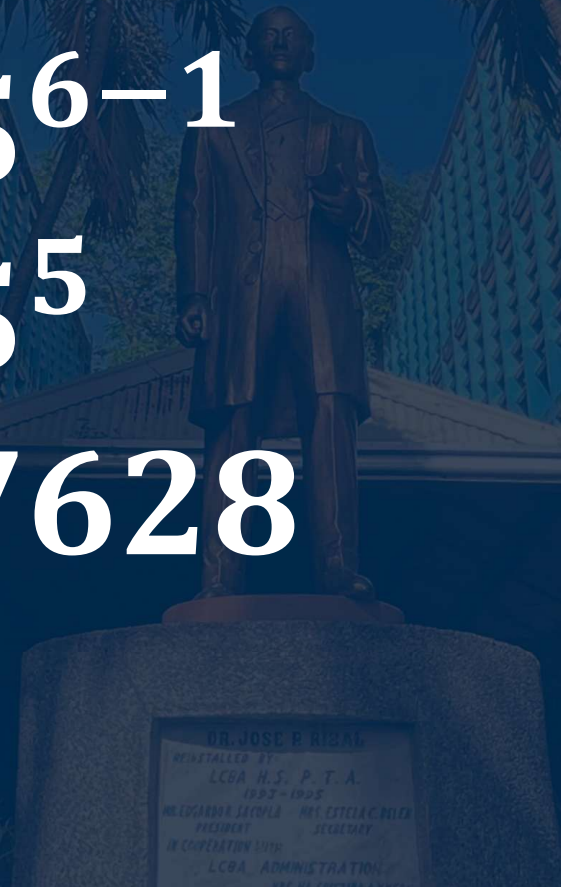
$$a_6 = 20,000 \cdot 1.05^{6-1}$$

$$a_6 = 20,000 \cdot 1.05^5$$

$$a_6 = 20,000 \cdot 1.27628$$

$$a_6 = 25,525.63$$

....





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Arithmetic Sequences in Real Life





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Illustrative Example 2: Weekly savings

You save ₱500 every week. Your total savings form an arithmetic sequence: 500, 1000, 1500, 2000, ... How much will you have saved after 10 weeks? After 25 weeks?





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Solution:

If $n = 10$, $d = 500$ and $a_1 = 500$,

$$a_n = a_1 + (n - 1)d$$

$$a_{10} = 500 + (10 - 1)500$$

$$a_{10} = 5000$$

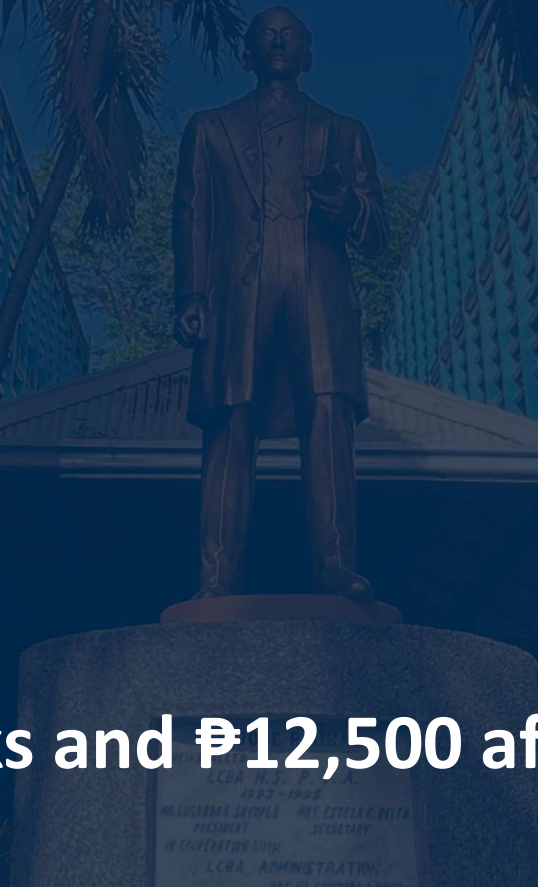
If $n = 25$, and $a_1 = 500$,

$$a_n = a_1 + (n - 1)d$$

$$a_{10} = 500 + (25 - 1)500$$

$$a_{10} = 12,500$$

Thus, you can save ₱5,000 after 10 weeks and ₱12,500 after 25 weeks.

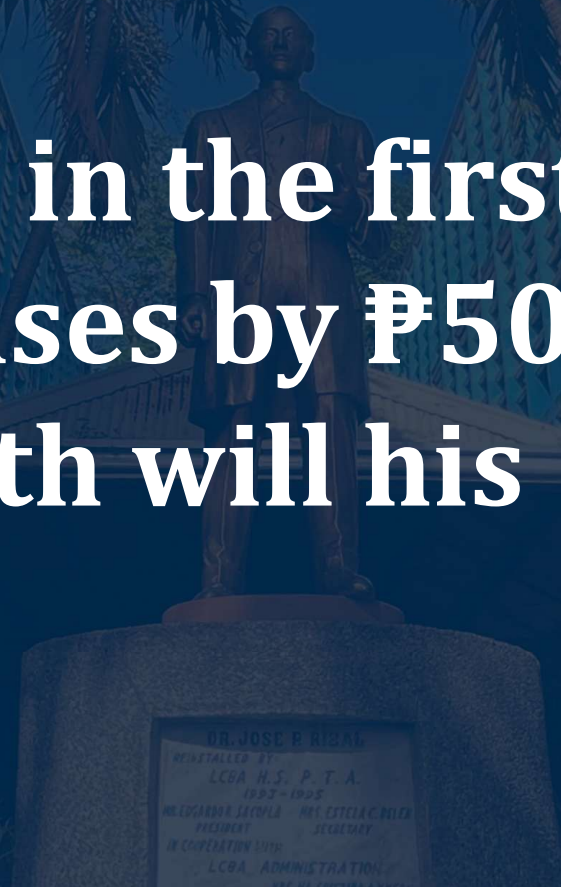




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Illustrative Example 3: Salary increments

A worker earns ₱12,000 in the first month. His salary increases by ₱500 every month. What month will his salary reach ₱15,000?





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If $a_1 = 12,000$, $d = 500$, and $a_n = 15,000$

$$a_n = a_1 + (n - 1)d$$

$$15,000 = 12,000 + (n - 1)500$$

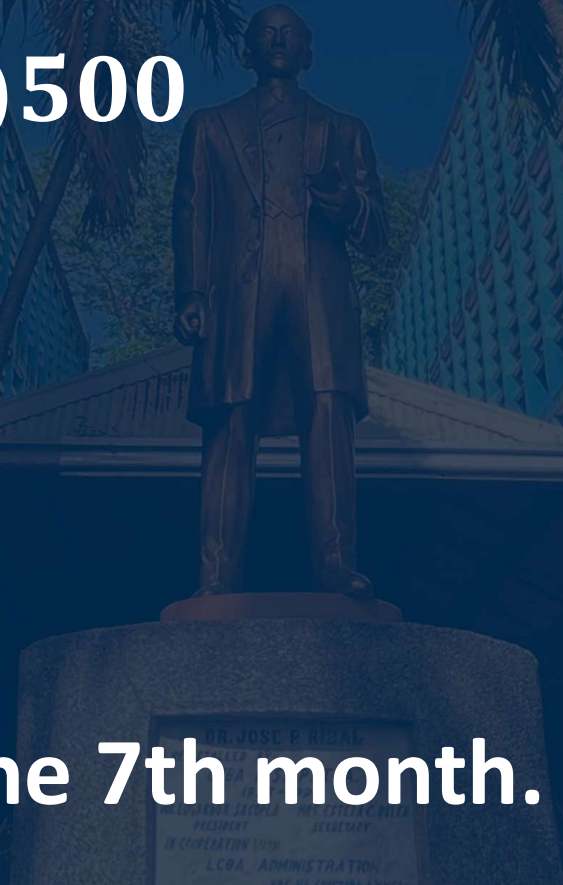
$$3,000 = (n - 1)500$$

$$n - 1 = \frac{3,000}{500}$$

$$n - 1 = 6$$

$$n = 6$$

His salary will reach ₱15,000 in the 7th month.





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Arithmetic sequences model situations with fixed, regular increments. Installment payments are a classic example.





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Illustrative Example 4: Gadget purchase:

You bought a smartphone costing ₱18,000 with a 6-month installment plan of ₱3,000 per month. How much will you pay on the 4th month?





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Solution:

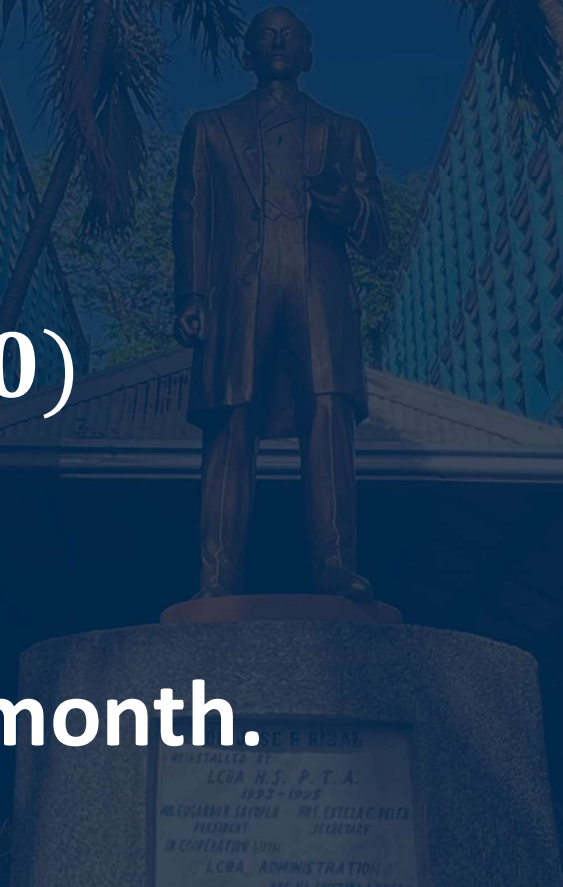
If $n = 4$, $d = 3,000$ and $a_1 = 3,000$,

$$a_n = a_1 + (n - 1)d$$

$$a_4 = 3,000 + (4 - 1)(3,000)$$

$$a_n = 12,000$$

You will pay ₱12,000 on the 4th month.





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Illustrative Example 5: School tuition

A student pays ₱2,500 per month for 10 months for his school tuition. What is the total paid by the 10th month?

.....



Solution:

If $n = 10$, $d = 2,500$ and $a_1 = 2,500$

$$a_n = a_1 + (n - 1)d$$

$$a_{10} = 2,500 + (10 - 1)(2,500)$$

$$a_{10} = 25,000$$

The total paid by the end of the 10th month is ₱25,000.



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Illustrative Example 6: Car loan

The monthly payment for a car loan is ₱7,500 for 36 months.
How much will be paid after 2 years (24 months)?

....



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Solution:

If $n = 24$, $d = 7,500$ and $a_1 = 7,500$

$$a_n = a_1 + (n - 1)d$$

$$a_{24} = 7,500 + (24 - 1)(7,500)$$

$$a_{24} = 180,000$$

₱180,000 will be paid after 2 years.





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Geometric Sequences in Real Life





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Geometric sequences describe multiplicative growth or decay, common in population studies, investments, and compound interest.





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Illustrative Example 7: Bacteria population:

A culture starts with 100 bacteria and doubles every hour. How many bacteria after 5 hours?





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Solution:

If $n = 5$, $r = 2$ and $a_1 = 7,500$

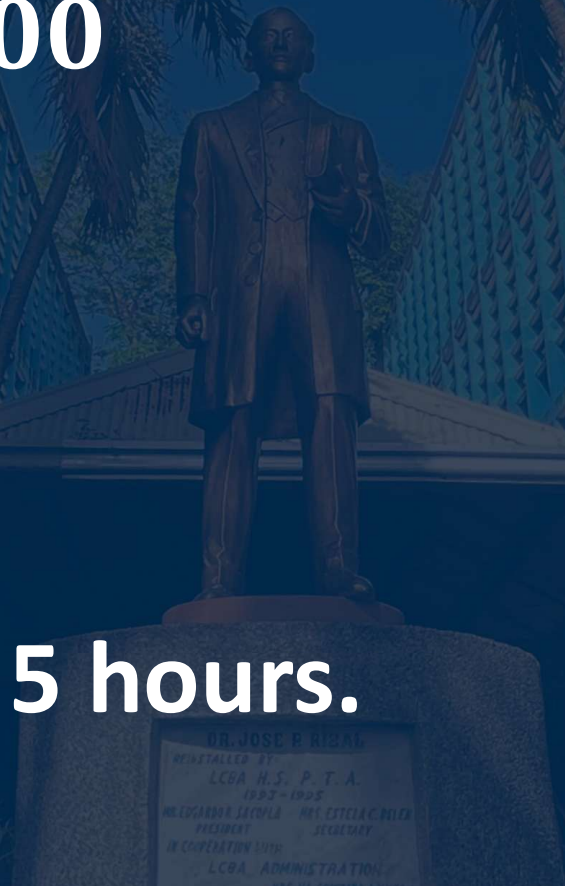
$$a_n = a_1 \cdot r^{n-1}$$

$$a_5 = 100 \cdot 2^{5-1}$$

$$a_5 = 100 \cdot 2^4$$

$$a_5 = 1,600$$

There are 1,600 bacteria after 5 hours.





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Illustrative Example 8: Town population

A town has 10,000 residents, growing at 5% per year.

Population after 3 years?

.....



Solution:

If $n = 3$, $r = 1 + 0.05 = 1.05$ and $a_1 = 10,000$

$$a_n = a_1 \cdot r^{n-1}$$

$$a_3 = 10,000 \cdot (1.05)^{3-1}$$

$$a_3 = 10,000 \cdot 1.05^2$$

$$a_3 = 11,025$$

After 3 years, the population is 11,025.





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Illustrative Example 9: Startup user base:

A mobile app gains 1,000 users the first week and doubles each subsequent week. How many users by the 6th week?



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Solution:

If $n = 6$, $r = 2$ and $a_1 = 1,000$

$$a_n = a_1 \cdot r^{n-1}$$

$$a_6 = 1,000 \cdot 2^{6-1}$$

$$a_6 = 1000 \cdot 2^5$$

$$a_6 = 32,000$$

The mobile app will have 32,000 users by the 6th week.



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1982-1985
MARGARITA JACULA PRESIDENT
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Depreciation reduces value over time and can be modeled as a geometric sequence.





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Illustrative Example 10: Gadget depreciation

A laptop costs ₱60,000 and loses 15% of its value each year. What is its value after 4 years?





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Solution:

If $n = 4$, $r = 1 - 0.15 = 0.85$ and $a_1 = 60,000$

$$a_n = a_1 \cdot r^{n-1}$$

$$a_4 = 60,000 \cdot 0.85^{4-1}$$

$$a_4 = 60,000 \cdot 0.85^3$$

$$a_4 = 36,847.50$$

The laptop will only cost ₱36,847.50 after 4 years.



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Illustrative Example 11:
Electronic gadget
A television worth ₱30,000
depreciates 20% per year.
What is its value after 3 years?

.....



Solution:

If $n = 3$, $r = 1 - 0.20 = 0.80$ and $a_1 = 30,000$

$$a_n = a_1 \cdot r^{n-1}$$

$$a_3 = 30,000 \cdot 0.8^{3-1}$$

$$a_3 = 30,000 \cdot 0.8^2$$

$$a_3 = 19,200$$

The value of the television after 3 years is



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Illustrative Example 12: Investment Growth

John invested ₱1,000 in a fund. After one year, the investment grew to ₱1,200. What is the common ratio (r) of the investment growth?





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Solution:

If $a_1 = 1,000$ and $a_2 = 1,200$

$$a_n = a_1 \cdot r^{n-1}$$

$$a_2 = 1,000 \cdot r^{2-1}$$

$$1,200 = 1,000 \cdot r^1$$

$$r = \frac{1200}{1000}$$

$$r = 1.2$$

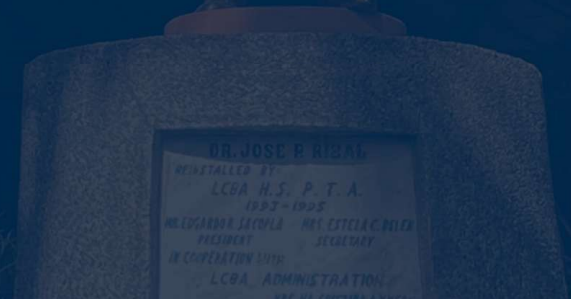
The investment grows by a factor of 1.2 each year (or 20% per year).





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MATH DRILL: PATTERNS IN LIFE AND NATURE





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Directions: Identification and Computation. Identify the type of sequence and the appropriate formula to use. Show your complete solution and round your final answer to two (2) decimal places.





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1. You save ₱500 every month. How much will you have after 12 months?





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2. The population of rabbits starts at 50 and doubles every month. How many rabbits after 6 months?



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3. A phone costs ₱20,000 and loses 10% of its value each year. What is its value after 5 years?

.....



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4. You pay a monthly installment of ₱2,000 for 18 months. How much will you have paid after 10 months?



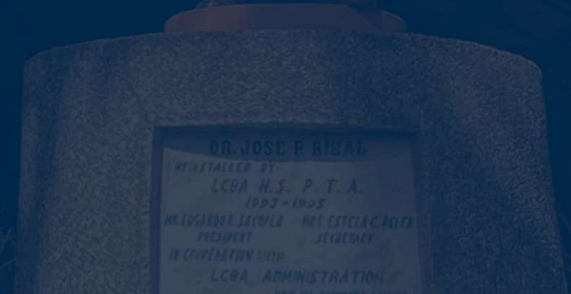
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5. A small investment grows by 5% each month starting at ₱1,000. How much is it worth after 4 months?



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GENERALIZATION: LESSON WRAP UP!





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- ✓ Today, we learned that arithmetic and geometric sequences are powerful tools for understanding patterns in numbers and in real life. Arithmetic sequences show steady growth through constant addition, such as savings, salaries, and installment payments. Geometric sequences, on the other hand, model rapid growth or decay through constant multiplication, such as investments, population changes, and depreciation of gadgets.



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✓ We also learned the formulas for computing terms in these sequences:

Arithmetic: $a_n = a_1 + (n - 1)d$

Geometric: $a_n = a_1 \cdot r^{n-1}$





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- ✓ Through the lesson, we realized that recognizing these patterns helps us plan ahead, budget wisely, and make informed decisions. Arithmetic sequences teach us discipline in handling fixed increments, while geometric sequences show us how growth and decline can affect our future. By applying these concepts, we can become more responsible students, employees, consumers, and entrepreneurs who use mathematical thinking to manage resources, predict outcomes, and contribute positively to society.

